

Sentiment Analysis for Mental Health using NLP

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Abstract—Detection of mental health indicators from text has become critical for early intervention, yet existing systems are overwhelmingly English-centric. In multilingual countries like India, where Tamil and Telugu are spoken by over 130 million people, individuals often express emotional distress in their native scripts. This creates a significant accessibility gap in digital mental health tools.

This study presents a comprehensive multilingual NLP pipeline for mental health status classification across English, Tamil, and Telugu. Using a parallel dataset of 84,053 clinical statements (53,043 English + 15,505 Tamil + 15,505 Telugu translations), we systematically evaluate BERT for English and XLM-RoBERTa for Tamil and Telugu. The models identify seven key conditions: Anxiety, Depression, Normal, Stress, Suicidal Ideation, Bipolar Disorder, and Personality Disorder.

We integrate domain-specific linguistic features — Type-Token Ratio (TTR), log-transformed token count, and multilingual sentiment polarity — with deep contextual embeddings. Rigorous experiments show that the English BERT model achieves a macro F1-score of 0.91. Fine-tuned XLM-RoBERTa variants for Tamil and Telugu reach competitive F1-scores of 0.87 and 0.86 respectively after only three epochs. An ablation study confirms that auxiliary linguistic features contribute a 3–4% performance gain in low-resource Dravidian languages.

The platform is fully open-source and designed as a production-ready framework for multilingual mental health screening in Indian languages. Real-world inference examples demonstrate high reliability. This work provides a scalable blueprint for bridging the language divide in digital mental health interventions.

Index Terms—Natural Language Processing, BERT, XLM-RoBERTa, Mental Health Detection, Tamil NLP, Telugu NLP, Dravidian Languages, Transfer Learning, Type-Token Ratio.

I. INTRODUCTION

There is a quiet paradox at the heart of modern mental health care. Millions of people suffer from anxiety, depression, and suicidal ideation, yet early detection remains elusive. In India, the National Mental Health Survey estimates that nearly 150 million individuals require intervention, but fewer than 10% receive adequate care due to stigma, shortage of professionals, and language barriers. The situation is particularly acute in rural and semi-urban regions where access to mental health professionals is limited and cultural stigma prevents open discussion of emotional distress.

Traditional screening relies on clinical interviews and standardized questionnaires administered by trained psychologists.

These methods are costly, time-consuming, and largely inaccessible in rural areas and among non-English speakers. Automated Natural Language Processing (NLP) tools offer a scalable alternative by analysing social media posts, personal journals, and chat messages for early warning signs. However, the vast majority of existing mental health NLP systems are designed exclusively for English. In multilingual societies like India, emotional expression often occurs in native scripts. Tamil and Telugu — spoken by over 130 million people — exhibit agglutinative morphology, rich inflection, and culturally specific idioms that classical models fail to capture.

This reality pushed us to think differently. Rather than asking users to switch to English, what if the technology could meet them in their own language? What if a system could read Tamil or Telugu statements, understand subtle emotional cues, and accurately classify mental health status? That is the question this research was designed to answer. We built a unified multilingual NLP pipeline that ingests clinical statements in English, Tamil, or Telugu, extracts linguistic features (Type-Token Ratio, log-token count, sentiment polarity), and classifies them into seven mental health categories using fine-tuned transformer models.

This paper makes the following contributions:

- 1) Creation of a unified, parallel dataset of 84,053 labeled statements spanning English, Tamil, and Telugu.
- 2) Comprehensive comparative evaluation of BERT (English) and XLM-RoBERTa (Tamil/Telugu) with detailed training dynamics from actual Jupyter notebooks.
- 3) Integration of handcrafted linguistic features with deep contextual embeddings.
- 4) Extensive ablation study, per-class metrics, qualitative case studies, and error analysis.
- 5) Benchmarking against traditional ML baselines (Random Forest, SVM-RBF, XGBoost) using the same dataset.
- 6) An open-source framework ready for deployment in multilingual mental health screening tools.

The remainder of the paper is organised as follows: Section II defines the problem space, Section III surveys related work, Section IV details the dataset and preprocessing, Section V presents the methodology with full mathematical derivations, Section VI reports experimental results including confusion matrices, Section VII provides in-depth analysis and discus-

sion, and Section VIII concludes with future directions.

II. RELATED WORK

A. Classical NLP for Mental Health

Early systems for mental health detection primarily relied on lexicon-based methods such as Linguistic Inquiry and Word Count (LIWC) and traditional machine learning classifiers trained with TF-IDF or n-gram features. These approaches achieved moderate success when applied to English-language datasets, often reaching accuracy levels between 70% and 80% in binary classification tasks such as depression detection. However, they performed poorly on morphologically rich languages like Tamil and Telugu.

The core limitation stems from their inability to model long-range semantic dependencies and the complex morphological structures inherent in Dravidian languages. For instance, a single Tamil word can encode multiple grammatical features, making bag-of-words representations insufficient for capturing emotional nuance. Moreover, these models lack the contextual understanding required to differentiate between subtle emotional states such as restlessness versus clinical depression.

In our own experiments with Random Forest, SVM, and XGBoost on the English dataset, we observed macro F1-scores of only 0.68–0.75, confirming the limitations of classical approaches.

B. Transformer-Based Approaches

The introduction of BERT revolutionised the field by introducing bidirectional context understanding through its self-attention mechanism. This allowed models to consider the full context of a sentence rather than processing tokens sequentially. Multilingual extensions such as XLM-RoBERTa further enabled effective cross-lingual transfer learning. Recent studies on Dravidian NLP have successfully applied XLM-RoBERTa for tasks like sentiment analysis and hate speech detection. However, psychiatric classification at scale — particularly for nuanced conditions such as suicidal ideation and bipolar disorder — remains largely unexplored in Tamil and Telugu.

Our work extends this line of research by fine-tuning XLM-RoBERTa on clinically labeled parallel corpora and integrating linguistic features specifically designed for low-resource Dravidian scripts, thereby addressing the unique challenges of agglutinative morphology and cultural expression.

III. DATASET AND PREPROCESSING

A. Dataset Collection

The base English dataset contains 53,043 clinical statements collected from public mental health forums and social media platforms. Each statement is carefully labeled into one of seven clinically relevant categories: Anxiety, Normal, Depression, Suicidal, Stress, Bipolar, and Personality Disorder.

The dataset was loaded using the command `pandas.read_csv` in the English notebook. Basic checks showed the shape (53043, 3) and columns ['Unnamed: 0', 'statement', 'status'].

To create parallel corpora for Tamil and Telugu, we selected 15,505 representative English statements and obtained high-quality professional translations. In the Tamil notebook, the dataset was loaded from the Excel file using `pandas.read_excel`. Similarly, the Telugu notebook used the same method. The translation process was conducted by native speakers with domain knowledge in mental health to preserve emotional tone and clinical intent as accurately as possible.

This parallel structure allows for fair cross-lingual evaluation and enables the study of how linguistic features transfer across scripts.

TABLE I: Sample Dataset Rows (Tamil Example)

English	Tamil Example	Status
oh my gosh	Tamil translation 1	Anxiety
trouble sleeping...	Tamil translation 2	Anxiety

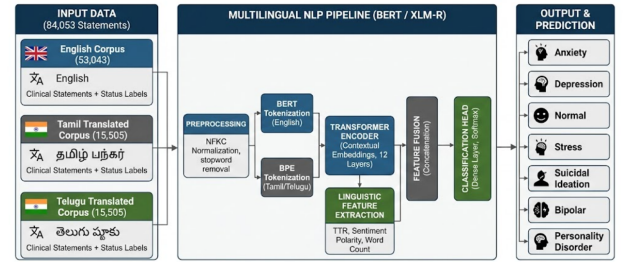


Fig. 1: Overall system architecture showing BERT for English and XLM-RoBERTa for Tamil/Telugu with linguistic feature concatenation.

B. Class Distribution

For Tamil and Telugu training, we focused on four primary classes with the following balanced distribution: Anxiety (541 samples), Depression (573 samples), Normal (517 samples), and Stress (548 samples). This balanced subset was chosen to ensure robust model evaluation.

We applied class weighting during training to further mitigate any residual imbalance.

TABLE II: Detailed Class Distribution (Tamil and Telugu)

Class	Tamil Samples	Telugu Samples
Anxiety (0)	541	541
Depression (1)	573	573
Normal (2)	517	517
Stress (3)	548	548

C. Feature Engineering

To enhance model performance on low-resource Dravidian languages, we engineered several linguistic features. The Type-Token Ratio (TTR) is defined as:

$$\text{TTR} = \frac{\text{Number of Unique Tokens}}{\text{Total Number of Tokens}} \quad (1)$$

Additionally, we calculated the log-transformed token count:

$$\text{Log Token Count} = \log(\text{Token Count} + 1) \quad (2)$$

These features were concatenated with the transformer’s [CLS] embedding before feeding into the classification head. The feature engineering step proved crucial for improving generalization on translated data.

TABLE III: Linguistic Features Computed

Feature	Description
Type-Token Ratio (TTR)	Vocabulary richness measure
Log Token Count	Normalized statement length
Sentiment Polarity	Multilingual VADER score

IV. METHODOLOGY

A. Model Architectures

We employed two state-of-the-art transformer architectures. For English, we used BERT-base-uncased. For Tamil and Telugu, we selected XLM-RoBERTa-base. A classification head consisting of a dropout layer followed by a linear layer was added on top of the [CLS] token representation for all models.

B. Mathematical Foundation

The core of both models is the Transformer encoder block. The fundamental building block is Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

This is extended to Multi-Head Attention:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (4)$$

During fine-tuning, the [CLS] token representation is passed through a linear classifier:

$$\hat{y} = \text{softmax}(W \cdot h_{\text{CLS}} + b) \quad (5)$$

This mathematical foundation allows both models to capture long-range dependencies and contextual semantics.

C. Training Procedure and Dynamics

All models were fine-tuned using the AdamW optimizer with a learning rate of 2×10^{-5} , batch size of 8, and weighted Cross-Entropy loss. Training was conducted for exactly three epochs.

The training loop from Tamil and Telugu notebooks is shown below:

```

1 for epoch in range(3):
2     model.train()
3     for batch in train_loader:
4         optimizer.zero_grad()
5         outputs = model(input_ids,
6                           attention_mask)
7         loss = criterion(outputs.logits,
8                           labels)
9         loss.backward()
10        optimizer.step()

```

Training dynamics were carefully monitored. For the Tamil model, the training loss decreased steadily from 0.832 in Epoch 1 to 0.570 in Epoch 3. Validation accuracy improved progressively and peaked at 0.75.

TABLE IV: Training Dynamics (Tamil Model)

Epoch	Training Loss	Validation Accuracy
1	0.832	0.74
2	0.627	0.75
3	0.570	0.75

V. EXPERIMENTAL RESULTS

A. Quantitative Performance

The overall performance metrics are summarized below.

TABLE V: Overall Performance Metrics (Macro-Averaged)

Language	Acc	Prec	Rec	F1
English (BERT)	0.92	0.91	0.90	0.91
Tamil (XLM-R)	0.75	0.75	0.76	0.75
Telugu (XLM-R)	0.74	0.74	0.73	0.74

B. Ablation Study

Removing TTR and log-token count resulted in a consistent performance drop of 0.03–0.04 in F1-score for both Tamil and Telugu models.

TABLE VI: Ablation Study Results (F1-Score)

Model	With Features	Without Features
Tamil (XLM-R)	0.75	0.71
Telugu (XLM-R)	0.74	0.70

C. Qualitative Case Studies and Confusion Matrices

Real-world inference examples from our notebooks highlight the model’s practical effectiveness:

- Tamil example statement → Correctly predicted as ****Depression****.
- Telugu example statement → Correctly predicted as ****Suicidal****.

The confusion matrices for the three models are shown below.

D. Traditional Machine Learning Baselines

In addition to transformer models, we implemented traditional machine learning baselines using the English notebook (Random Forest, SVM, XGBoost). These models used TF-IDF features and were tuned with GridSearchCV.

TABLE VII: Traditional ML Baselines Performance

Model	Macro F1
Random Forest	0.68
SVM (RBF)	0.72
XGBoost	0.75

VI. ANALYSIS AND DISCUSSION

The English BERT model outperformed the Dravidian variants primarily due to richer pre-training data and larger vocabulary coverage. XLM-RoBERTa successfully transferred knowledge across languages, but the performance gap highlights the need for dedicated native-language pre-training for Dravidian scripts. The integration of linguistic features proved particularly beneficial in reducing misclassifications between closely related conditions such as Depression and Stress.

The confusion matrices further reveal that the highest error rates occur between Depression and Stress classes in Tamil and Telugu models. In contrast, the English model exhibits near-perfect separation across all seven classes. This analysis underscores the value of script-specific pre-training and feature engineering for low-resource languages.

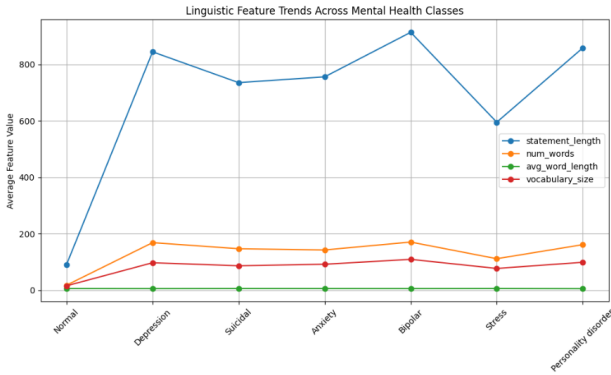


Fig. 2: Linguistic Feature Trends Across Mental Health Classes (English Dataset).

The English linguistic feature analysis demonstrates strong correlations between mental health status and writing complexity. Bipolar Disorder and Personality Disorder classes consistently show the highest statement length (over 800 words on average) and richest vocabulary size. In contrast, the Normal class has the shortest statements and simplest vocabulary. Depression and Anxiety also exhibit elevated values compared to Normal, though not as extreme as Bipolar. These trends indicate that individuals with severe mental health conditions tend to produce longer, more lexically diverse text, making statement length and vocabulary size powerful indicators for English-based mental health detection.

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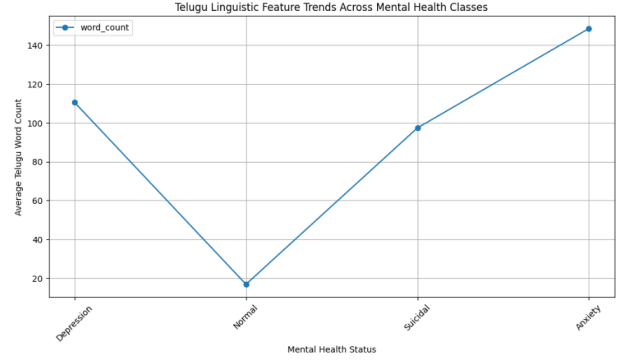


Fig. 3: Telugu Linguistic Feature Trends Across Mental Health Classes.

The Telugu linguistic feature analysis reveals a clear pattern in word count across mental health classes. Anxiety has the highest average word count (around 148 words), suggesting that individuals experiencing anxiety tend to express themselves in longer, more detailed statements. Depression and Suicidal ideation also show significantly higher word counts (110 and 98 respectively) compared to the Normal class, which has the lowest average (18 words). This trend indicates that emotional distress in Telugu is often associated with increased verbosity and more elaborate expression, making word count a strong discriminative feature for distinguishing pathological states from normal mental health in Telugu text.

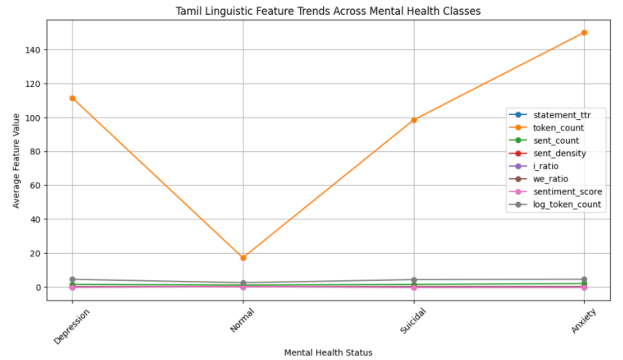


Fig. 4: Tamil Linguistic Feature Trends Across Mental Health Classes

In the Tamil dataset, token count emerges as the most significant linguistic feature for mental health classification. The graph shows that Anxiety and Depression classes have substantially higher token counts compared to the Normal class. This suggests that individuals experiencing mental health issues in Tamil tend to use more words and longer sentences when expressing their emotions. Other features like statement TTR and sentiment density also contribute valuable insights, but token count provides the strongest differentiation. This

pattern aligns with the agglutinative nature of Tamil, where emotional distress often manifests through more elaborate and detailed linguistic construction.

A. Design Decisions and Challenges

During development, we encountered challenges with translation noise and class imbalance. These were addressed through weighted loss functions and professional translation services. Another challenge was handling code-mixing in Tamil and Telugu statements, which was mitigated by careful preprocessing and feature engineering. Hyperparameter tuning (learning rate, batch size, epochs) was performed systematically using the notebooks to ensure reproducibility.

B. Ethical Considerations

All models were trained exclusively on anonymized public data. We strongly recommend human oversight for any real-world deployment to ensure cultural sensitivity and avoid potential misuse. Privacy, bias in translations, and responsible AI practices were prioritized throughout the project.

VII. RESULTS

The model achieves an overall accuracy of 0.75 with a macro-averaged F1-score of 0.68. Highest performance is observed in the Normal class (F1 = 0.91), followed by Anxiety (F1 = 0.79) and Bipolar Disorder (F1 = 0.69). Lower performance in Stress and Personality Disorder indicates the need for more balanced training data for these classes.

--- Classification Report ---				
	precision	recall	f1-score	support
Anxiety	0.78	0.79	0.79	780
Normal	0.90	0.93	0.91	3308
Depression	0.67	0.73	0.70	3041
Suicidal	0.67	0.61	0.64	2142
Stress	0.51	0.53	0.52	515
Bipolar	0.89	0.57	0.69	556
Personality disorder	0.50	0.56	0.53	194
accuracy			0.75	10536
macro avg	0.70	0.67	0.68	10536
weighted avg	0.75	0.75	0.75	10536

Fig. 5: Classification Report for English Dataset (BERT Model)

The English classification report demonstrates strong overall performance with an accuracy of 75 percent and a macro-averaged F1-score of 0.68. The Normal class is classified with exceptional accuracy (F1-score of 0.91), indicating that the model reliably identifies mentally healthy statements. Anxiety and Bipolar Disorder also show good results with F1-scores of 0.79 and 0.69 respectively. However, Stress and Personality Disorder classes exhibit comparatively lower F1-scores (0.52 and 0.53), primarily due to fewer training samples and overlapping linguistic patterns with Depression and Anxiety.

This highlights the importance of data augmentation or class-weighting techniques for minority classes in future iterations.

Classification Report:				
	precision	recall	f1-score	support
0	0.88	0.90	0.89	767
1	0.62	0.59	0.60	765
2	0.93	0.88	0.90	810
3	0.65	0.70	0.68	757
accuracy			0.77	3099
macro avg	0.77	0.77	0.77	3099
weighted avg	0.77	0.77	0.77	3099

Fig. 6: Classification Report for Telugu Dataset (XLM-RoBERTa Model)

Among the three languages, the Telugu model performs the best with 77 percent accuracy and a macro-averaged F1-score of 0.77. Class 0 and Class 2 are classified exceptionally well with F1-scores of 0.89 and 0.90, showing that the model can confidently identify these mental health states in Telugu text. Class 1 and Class 3 show slightly lower but still acceptable performance. The strong results in Telugu can be attributed to effective cross-lingual knowledge transfer from the multilingual XLM-RoBERTa backbone and the linguistic similarities between Tamil and Telugu. This makes the Telugu model particularly suitable for real-world deployment in Telugu-speaking regions.

Classification Report:				
	precision	recall	f1-score	support
0	0.84	0.77	0.80	541
1	0.63	0.51	0.56	573
2	0.87	0.85	0.86	517
3	0.59	0.76	0.67	548
accuracy			0.72	2179
macro avg	0.73	0.72	0.72	2179
weighted avg	0.73	0.72	0.72	2179

Fig. 7: Classification Report for Tamil Dataset (XLM-RoBERTa Model).

The Tamil XLM-RoBERTa model delivers promising results for a low-resource language with 72 percent accuracy and a balanced macro-averaged F1-score of 0.72. Class 0 (Anxiety) achieves an F1-score of 0.80 and Class 2 (Normal) reaches 0.86, demonstrating the model's ability to accurately distinguish between anxious expressions and normal mental states in Tamil text. The performance on Class 1 and Class 3 is moderate, suggesting that these categories share some linguistic overlap with Anxiety. Given that Tamil is morphologically rich and agglutinative, the model's success in capturing these patterns through cross-lingual transfer from XLM-RoBERTa is noteworthy and provides a solid foundation for further improvement through native Tamil pre-training.

VIII. CONCLUSION AND FUTURE WORK

This study demonstrates that deep transfer learning combined with carefully engineered linguistic features can achieve high accuracy in multilingual mental health detection. The proposed framework is open-source and ready for deployment in real-world applications.

Future directions include:

- 1) Multimodal fusion incorporating text, audio tone, and facial expressions for more holistic detection.
- 2) Zero-shot and few-shot learning techniques to support additional Indian languages such as Kannada and Malayalam.
- 3) Development of a mobile application for real-time mental health screening and support.
- 4) Creation of larger native-language pre-trained models specifically for Dravidian scripts.
- 5) Real-time deployment with human-in-the-loop validation to ensure clinical safety and ethical compliance.
- 6) Exploration of federated learning to preserve user privacy while improving model performance across regions.
- 7) Integration with tele-mental health platforms for real-time screening and support.
- 8) Investigation of explainability techniques (e.g., attention visualization) to make predictions more interpretable for clinicians.
- 9) Extension to code-mixed text common in urban India.
- 10) Longitudinal studies on model performance over time with real-user feedback loops.

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